

EventFlow

Event Scheduling & Participation Platform

Project Documentation

Introduction

EventFlow is a modular event scheduling and participation platform designed to help organizations create, configure, and operate multiple independent events through a centralized system.

Each event functions as an isolated workspace with its own administrators, staff, participants, schedules, registrations, attendance, announcements, dashboards, media, and event-specific settings.

The platform provides two primary experiences: an event management experience for authorized event members and a participant experience that makes event timing, activities, venues, notifications, attendance, photographs, and certificates easy to understand and access.

The system can also support advanced capabilities such as automated participant-photo identification, personalized galleries, analytics, feedback, and configurable event workflows.

Problem Statement

Organizations often manage events using a combination of spreadsheets, registration forms, messaging applications, calendars, attendance tools, and separate photo-sharing systems. This creates fragmented workflows and makes it difficult to maintain a consistent view of an event.

Common problems include manual schedule coordination, participant confusion about event timing, conflicts between sessions and venues, unclear responsibilities among event staff, difficulty tracking registration and attendance, limited event-specific dashboards, manual communication of schedule changes, and difficult post-event photo distribution.

The problem becomes more complex when several events are managed at the same time because data, permissions, and responsibilities must remain isolated between events.

Existing Approach and Limitations

Existing event-management approaches commonly provide event creation, registration, ticketing, scheduling, notifications, attendance, and basic reporting. However, many implementations are centered on generic event workflows and may provide limited support for strict event isolation, section-level responsibilities, personalized participant schedules, configurable dashboards, and post-event participant photo discovery.

The proposed system addresses these limitations by treating every event as an independent operational workspace while still allowing the same person to participate in multiple events with different responsibilities.

Proposed System

EventFlow proposes a centralized platform in which every event operates independently. An event can define its own administrators, sections, staff assignments, participants, registration process, schedule, venues, attendance rules, announcements, dashboards, photographs, certificates, feedback, and settings.

Users may participate in multiple events, but event-level permissions are evaluated in the context of the selected event and, when required, the selected section or session.

The platform therefore supports the model: User + Event + Role + Scope = Permission.

Event Isolation

Event isolation is a core design principle. Data and operational actions belonging to one event must not interfere with another event.

```
Platform
|-- Event A
|   |-- Admins
|   |-- Staff
|   |-- Participants
|   |-- Schedule
|   |-- Attendance
|   |-- Photos
|   `-- Dashboard
|-- Event B
|   -- ...
-- Event C
   -- ...
```

Changes made inside Event A, such as schedule updates, photo uploads, registration closure, or dashboard configuration, must not modify Event B or Event C.

Role and Scope Model

The platform separates platform-level authority from event-level authority. Most operational roles are assigned to a specific event and can optionally be restricted to a section, session, venue, or other operational scope.

Role	Scope	Purpose
Platform Administrator	Entire platform	Manages organizations, platform users, global configuration, security, and system-level administration.
Event Owner	Specific event	Owns the event, controls its lifecycle, configures major settings, and assigns event administrators and staff.
Event Administrator	Specific event	Manages the complete operation of the selected event, including participants, schedules, registrations, announcements,

		attendance, dashboards, photos, certificates, feedback, and settings.
Schedule Manager	Event / section / session	Creates and manages activities, sessions, time slots, venues, speakers, schedule changes, and conflict detection.
Registration Manager	Event / registration section	Configures registration, forms, tickets, capacity, waitlists, approvals, cancellations, and participant information.
Attendance Staff	Event / section / session	Performs check-in/check-out, QR verification, session attendance, and attendance corrections within the assigned scope.
Photographer	Event / section / session	Uploads, organizes, and manages photographs for the assigned event area and supports photo-processing and participant matching workflows.
Volunteer / Event Staff	Event / section	Performs assigned operational activities such as participant assistance, venue support, queue management, and session support.
Speaker	Event / session	Manages speaker information, views assigned sessions and venues, provides materials, and receives relevant schedule updates.
Sponsor	Specific event	Manages sponsor information, promotional materials, sponsorship details, and event-related visibility.
Participant	Specific event	Registers for the event, accesses tickets and schedules, receives notifications, participates in sessions, views attendance, photos, certificates, and feedback.

Section-Level Responsibilities

An event may contain multiple sections, activities, sessions, or operational areas. Responsibilities can be assigned at the event level or narrowed to a specific section or session. For example, an Event Administrator may control the whole event, a Photographer may be assigned only to the Cultural Program, and Attendance Staff may be responsible only for a particular workshop.

Event Scheduling

The scheduling module manages activities through structured time slots. Each activity can include a name, description, start time, end time, venue, speaker, capacity, section, and participation requirements. The system should detect time conflicts, venue conflicts, speaker conflicts, and participant schedule conflicts.

Participant Schedule

Participants should receive a personalized event timeline rather than being required to understand the entire event manually. The system can show registration, current activity, next activity, upcoming sessions, locations, and schedule changes relevant to the participant.

Event Dashboard

Every event can have its own dashboard containing event-specific information such as participant count, registration status, attendance, capacity, session participation, upcoming activities, trends, feedback, photo-processing status, and other metrics. Dashboard configuration belongs to the event.

Participant Dashboard

The participant dashboard provides a focused view of the selected event: registration, ticket, personal schedule, attendance, announcements, notifications, photos, certificate status, and feedback. A participant can switch between events without mixing the data of those events.

Event Communication

Each event has its own announcement and notification context. Administrators can communicate with all participants or targeted groups such as section participants, session participants, checked-in participants, or other defined groups. Schedule changes should notify only the affected audience.

Photo Management and Identification

Photographers upload photographs to the appropriate event or section. The system can process uploaded images, detect faces, match participants, and make relevant photographs available in a participant-specific gallery. Photos must remain associated with the correct event and scope.

Attendance Management

Attendance is maintained in the context of an event and can optionally be tracked at session level. The system can support event check-in, check-out, QR-based verification, session attendance, attendance correction, and attendance statistics.

Certificates

Each event can define its own certificate rules. Examples include participation certificates, completion certificates based on attendance requirements, and winner certificates for competitions. Certificate logic should be configurable per event.

Feedback and Evaluation

Participants can submit feedback about the event, sessions, speakers, venue, organization, and overall experience. Event administrators can use the collected feedback to evaluate the event and identify improvement areas.

Event Templates and Cloning

The platform can support reusable event templates and event cloning. A template may contain registration structure, schedule structure, dashboard configuration, notification templates, and certificate configuration. When an event is cloned, the resulting event becomes independent of the source event.

Multiple Roles for One User

A single user may have different roles in different events. For example, the same person can be an Event Administrator in one event, a Photographer in another, and a Participant in a third. Permissions are therefore evaluated using the user, selected event, assigned role, and scope.

Core Access Principle

No event-level role should automatically grant access to another event. Every request to event-specific information or action should be evaluated against the event context and the user's assigned scope.

Major Features

The major functional areas include independent event management, event-scoped roles, section-level permissions, scheduling and conflict detection, registration, custom forms, QR tickets, attendance, personalized participant schedules, announcements, notifications, configurable dashboards, photo management, AI-assisted participant photo identification, personal galleries, certificates, feedback, templates, cloning, and analytics.

Future Enhancements

Potential future capabilities include intelligent event recommendations, AI-assisted event planning, predictive attendance analysis, smart schedule optimization, mobile applications, real-time event updates, advanced photo search, networking, and gamification.

Conclusion

EventFlow provides a structured platform for managing multiple independent events within one application. Its defining principle is event isolation: each event maintains its own users, roles, responsibilities, schedules, registrations, attendance, photos, dashboards, communication, certificates, feedback, and settings. The participant experience is personalized to make event timing and participation easy to understand, while event administrators receive the operational control and analytics required to run each event effectively.